

Problem-SolutionFit

Date	6July2026
TeamID	
ProjectName	Plugging into the Future : An Exploration of Electricity consumption Patterns
MaximumMarks	5Marks

Problem-SolutionFit Canvas:

1. CUSTOMER SEGMENT(S)[CS] 1.Household consumers 2.Power distribution companies (DISCOMs) 3.Smart grid managers	6. CUSTOMER LIMITATIONS 1.Lack of energy-saving awareness 2.No smart-meter hardware access 3.Complex historical usage data	5. AVAILABLE SOLUTIONS Pros 1.Pros: Static monthly utility bills 2.Cons: No predictive foresight or breakdown of peak hours
2. PROBLEMS/PAINS [PR] 1.High energy wastage 2.Unpredictable power surges 3.Lack of real-time consumption insights	9. PROBLEMROOT/ CAUSE [RC] 1.Unmonitored appliance usage 2.Inaccurate baseline forecasting models	7. BEHAVIOR[BE] 1.Tracks daily peak usage 2.Adjusts high-load appliance times
3.TRIGGERS TO ACT [TR] 1.Rising electricity bills 2.Grid instability / Blackouts 3.Sustainability & carbon footprint goal 4. EMOTIONS Before - Anxious After - In Control	10. YOURSOLUTION [SL] 1.Machine Learning load forecasting 2.Flask energy monitoring dashboard 3.Pattern analysis & anomaly detection	8.CHANNELS OFBEHAVIOR [CH] ONLINE: Web analytics portal / Smart meter app OFFLINE: Direct consumer power panels

